

Physical AI Challenge – Vision-Based Localization and Autonomous Inspection

Physical AI combines robotics, computer vision, and intelligent reasoning to enable machines to understand and interact with the physical world. In this challenge, participants explore how AI can interpret real-world environments from video data to support autonomous inspection and infrastructure maintenance.

Challenge Overview

In sewer cleaning operations, a nozzle is connected to a high-pressure hose and inserted into a shaft. The redirected water flow generates a backward thrust that propels the nozzle through the pipe. A camera mounted on the nozzle records inspection videos while the system moves through the shaft.

The goal of this challenge is to develop a system that estimates the distance from the starting shaft directly from the video stream and displays this distance within the inspection video.

Currently, the turnaround point of the nozzle is estimated using the measured hose length and other indirect signals. Participants are invited to explore vision-based and AI-driven approaches that determine the traveled distance or detect the turnaround point in the video without adding additional hardware.

Participants may also extend their solution with AI-based inspection capabilities, such as classifying the cleanliness level of the pipe or detecting potential defects or anomalies within the pipe.

To address this challenge, participants are encouraged to explore a wide range of approaches, including classical computer vision and robotics localization techniques as well as transformer-based models, multimodal AI systems, and LLM-driven reasoning.

Bonus Ideas (Optional Inspiration)

Teams are free to explore any approach. The following ideas may serve as inspiration:

- Use visual odometry or SLAM techniques to estimate movement through the pipe
- Track visual features or patterns inside the pipe to infer distance traveled
- Use multimodal AI or foundation models to reason about video sequences
- Use LLMs to analyze frames or metadata to help detect the turnaround point
- Detect and highlight pipe conditions such as deposits, cracks, or root intrusion

These ideas are only suggestions. Creative and unconventional approaches are encouraged.

Provided Data

Participants will receive:

- Sample inspection videos
- Test and training videos with ground-truth distance annotations and turnaround point
- Metadata describing the inspection runs
- Assumed Skills

Required skills

Participants should ideally have experience in

- Computer vision (e.g., OpenCV, feature tracking, optical flow)
- Familiarity with machine learning, multimodal AI, or SLAM (beneficial but not required)
- Python-based prototyping and basic video processing

Hardware Requirements

Participants should bring their own laptops with:

- Python environment
- Computer vision libraries (e.g., OpenCV)
- Optional GPU or access to cloud AI services

Goal

Teams should deliver a prototype that takes an inspection video as input and produces an annotated output video or visualization. The result should show the estimated distance from the starting shaft and may optionally include basic pipe cleanliness classification or defect detection, demonstrating how AI and computer vision can support automated infrastructure inspection.

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